ARRAYS

Q1) MARTIN NUMERALS:

Question:

DS	**Session**	Arrays	**Question Information**	Level 1 Challenge 21

Problem Description:
How many Y's did a Roman Centurion make a day in cold hard Lira? About a C's worth! Turns out, Martians gave Rome the idea for their number system. Use the conversion charts below to help translate some Martian numbers!

Note, that unlike the Roman Numerals, Martian Numerals reuse symbols to mean different values. B can either mean '1' or '100' depending on where it appears in the number sequence.

Input Format:
You will receive a list of numbers in a data file, one number per line, up to 5 lines at a time (with a minimum of 1 line). No number will exceed 1000, or be less than 1.

Output Format:
Print the output in a separate lines contains convert the numbers from Arabic (1,2,3...10...500...1000) to Martian (B,BB,BBB...Z...G...R) numerals.

PROGRAM:

Q1.c

Arrays > Q1.c > ...

```c
#include<stdio.h>
#include<stdlib.h>

int main(){
    int values[] = {1000,900,500,400,100,90,50,40,10,9,5,4,1};
    char symbols[][4] = {"R","BR","G","BG","B","ZB","P","ZP","Z","BK","W","BW","B"};

    int N = sizeof(values)/sizeof(values[0]);

int t;
scanf("%d",&t);
int numbers[t];

for(int i=0; i<t; i++){
    scanf("%d",&numbers[i]);
}

for(int j=0; j<t; j++){
    int n = numbers[j];
    for(int i=0; i<N; i++){
        while(n >= values[i]){
            printf("%s",symbols[i]);
            n = n - values[i];
        }
    }
    printf("\n");
}

return 0;
}
```

OUTPUT:

TEST CASE 1:

```
5
26
67
186
408
500
ZZWB
PZWBB
BPZZZWB
BGWBBB
G
```

TEST CASE 2:

```
5
161
627
816
401
501
BPZB
GBZZWBB
GBBBZWB
BGB
GB
```